MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FIFTH SEMESTER THEORY EXAMINATION MODEL QUESTION PAPER 2026

(2024 ADMISSION ONWARDS)

B.Sc (HONS) BIOLOGICAL TECHNIQUES & SPECIMEN PREPARATION

MG5DSCBTS300- DEVELOPMENTAL BIOLOGY

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from e ach course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer All Questions
Each question carries 1 mark

1. Developmental biology is the study of **[K] [1]**
   a) cell metabolism. b) formation and development of organism from zygote to adult
   c)classification of organisms d) ecology of population
2. The term Ontogeny refers to **[U][1]**
   a) Evolution of species b) development of an individual organism c) study of genes d) population growth
3. The process by which an organism develops from an unfertilized egg is called **[K] [2]**
   a) Fertilization b) Gastrulation c) organogenesis d) parthenogenesis
4. The experimental organizer in Spemann's experiment was **[U] [3]**
   a) Animal pole b) Dorsal lip of blastopore c) vegetal pole d) Neural tube
5. The cortical reaction helps prevent **[U] [2]**
   a) Polyspermy b) cleavage c) gastrulation d) Implantation
6. A scientist wants to produce genetically identical animals. Which technique would be most appropriate? **[U][3]**
   a) Embryo tranfer b) cross fertilization c) amniocentesis d) cloning
7. Damage to the mesoderm during gastrulation would most directly affect the development of? **[U][4]**
   a) Nervous tissue only b) Muscles and connective tissues c) Epidermis only d) Digestive epithelium only
8. The hollow embryonic stage formed after cleavage is called **[K][2]**
   a) Blastula b) Gastrula c) Neural tube d) morula
9. The main function of the notochord during development is to **[U][4]**
   a) Produce blood cells b) Induce development of the nervous system c) Form the epidermis d) Form the digestive tract
10. Gene action during development mainly determines? **[U][5]**
    a) Cell specialization b) Cleavage only c) Water balance d) Fertilization

[1 x 10= 10]

Part B

Short questions Type

Answer any 10 Questions

Each question carries 2 marks

11. What is parthenogenesis? Give examples [**K**] [2]
12. What is the significance of the cortical reaction? **[K] [2]**
13. Explain the process of regeneration **[K] [4]**
14. Differentiate between blastula and gastrula? **[K] [2]**
15. Why is the dorsal lip of the blastopore called the organizer? **[U][3]**
16. Describe the importance of prenatal diagnosis **[U] [3]**
17. What are the different types of cloning? **[K][4]**
18. Explain Spemanns constriction experiments. **[K] [3]**
19. What are the major benefits of stem cell research? **[K][3]**
20. Describe the pattern of cleavage in the frog embryo. **[K][4]**
21. Explain different morphological and anatomical changes take place in frog metamorphosis **[K][4]**
22. Define blastula. Explain different types of blastula. **[K][2]**

[2 x 10= 20]

Part C

Short Essay Type Questions

Answer any 5 Questions
Each question carries 4marks

23. With suitable diagrams explain various types of morphogenetic movements seen in fertilized egg **[U][2]**

24. Explain gene action in cell differentiation with special reference to *Drosophila* **[U] [5]**

25. Discuss the major theories proposed in the history of embryology. **[U][1]**

26. Discuss the benefits and ethical concerns associated with cloning**. [U] [4]**

27. Explain how prenatal diagnosis can be applied for the detection of genetic abnormalities. **[U][3]**

28. Give an account of hormonal control of human reproduction **[K] [2]**

29. Compare and analyse the major differences between the estrous and menstrual cycles. **[An][2]**

30. Explain the mechanism involved in fertilization. **[K][2]** **[4 x 5= 20]**